KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2023
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define corpus. Write a regular expression to find an answer for the following statement. (08) [CO1]
"A computer with 6 GHz and 2 TB storage within 500$".
b) What is lemmatization and stemming? How is lemmatization done? (06) [CO1]
c) How does Byte Pair Encoding (BPE) work for tokenization? Use the following training corpus to tokenize the test word "hugs". (14) [CO1]
"This is the hugging face course. It shows tokenizer algorithms".
d) Make a classification of the types of spelling errors and their solution approaches. (07) [CO1]

2. a) You have two strings *St*1 and *St*2. Develop an algorithm to go from *St*1 to *St*2 with minimum cost of edit. Use insert, delete cost = 1 and update cost = 2. Explain your algorithm using example as *St*1 = "INTENTION" and *St*2 = "EXECUTION". (15) [CO3]
b) Define language model. "Predicting something is difficult". How does *N*-gram language model predict next word? Explain with suitable example. (14) [CO2]
c) How is language model evaluated? (06) [CO2]

3. a) Define text classification. "Naïve Bayes is a form of generative classifier" – justify the statement. (08) [CO3]
b) How does Naïve Bayes learn? Using information from the following table, find the class of the test sentence. (12) [CO3]

	Document	Class
Training	The movie is amazing	Positive
	I do not like the film	Negative
	Great acting and engaging	Positive
	Terrible movie and not recommended	Negative
Test	The acting is superb	?

c) What is parsing in NLP? What are the differences between syntactic and semantic parsing? Explain. (05) [CO3]
d) How does CYK algorithm work? For the following grammar G, is "$baba$" in $L(G)$? (10) [CO3]

$$S \rightarrow AB|BC$$
$$A \rightarrow BA|a$$
$$B \rightarrow CC|b$$
$$C \rightarrow AB|a$$

4. a) "You have a true word w and a noisy channel that modifies w into w'". How can you find the w from w'? Explain. (10) [CO5]
b) How can you define a word as a vector? How can you measure the similarity between two words u and v? Show the similarity of the words from the following table. (15) [CO5]

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

c) What is absolute discounting? Explain what are its advantages. (10) [CO5]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is closed and open class of Parts of Speech (POS)? Explain with example. (05) [CO2]
   b) Explain different type of taggers and describe Hidden Markov Model (HMM) POS tagging using example. (10) [CO1]
   c) Explain Viterbi algorithm for POS tagging with example. (10) [CO1]
   d) How do conditional random fields enhance POS tagging through feature incorporation and why are features challenging to integrate into HMMs? (10) [CO1]

6. a) What is relation extraction? Explain this concept using pattern-based relation extraction and supervised learning algorithm. (10) [CO2]
   b) Explain semi-supervised relation extraction via bootstrapping. (08) [CO2]
   c) What are temporal lexical triggers? How do we detect events? (06) [CO2]
   d) How is the *tf-idf* score computed for terms in documents and how can it be used to calculate the cosine similarity between a query and documents using BM25? Demonstrate for the following dataset. (11) [CO5]

   Assume, $|d| = 4$, $|d_{avg}| = 3$, $k = 1.3$, $b = 0.75$.

   Query: sweet love
   Doc 1: Sweet sweet nurse! Love?
   Doc 2: Sweet sorrow
   Doc 3: How sweet is love?
   Doc 4: Nurse!

7. a) Explain the evaluation of information retrieval systems and describe IR-based factoid question answering with a diagram. (08) [CO5]
   b) How can BERT-based models be effectively trained and evaluated for answer span extraction in information retrieval-based question answering systems? (09) [CO5]
   c) What is entity linking? Explain the process of entity mention detection using mathematical formulas and a diagram. (08) [CO5]
   d) What is knowledge-based question answering? Describe Watson's question answering system with the help of a diagram. (10) [CO5]

8. a) How does attention mechanism work in machine translation and why is it important for improving translation quality? (09) [CO4]
   b) Explain hourglass metaphor in text-to-speech with proper diagram. (08) [CO4]
   c) What is the difference between tokens and embeddings? Explain the pipeline of creating embedding model. (08) [CO4]
   d) Why is beam search necessary? Explain beam search with example. (10) [CO2]